An experimental comparison of at-issueness diagnostics

Word count (including references, excluding abstract and supplements): JT: xx.xxx

Abstract At-issueness is a key concept in semantics and pragmatics, distinguishing the main point of an utterance from background information. However, there is no consensus on how at-issueness should be defined or diagnosed, given that different diagnostics for at-issueness align with different definitions and rely on different assumptions about properties of at-issue content. This paper presents an experimental comparison of five diagnostics of at-issueness, applied to the contents of non-restrictive relative clauses and the complements of clause-embedding predicates. The results are mixed: While some comparisons of the results of diagnostics reveal differences, suggesting that the diagnostics are not interchangeable, other comparisons suggest that different diagnostics yield consistent results. The paper develops several hypotheses about factors that may lead different diagnostics to give different results and cautions against concluding that differences between the results of two diagnostics means that they measure distinct underlying concepts (cf., Snider 2017b; Koev 2018).

Keywords: definitions of at-issueness, properties of at-issue content, at-issueness diagnostics, experimental pragmatics, non-restrictive relative clauses, clause-embedding predicates

1 Introduction

At-issueness is a key concept in theoretical semantics and pragmatics,[1] used in the analysis of various phenomena including presupposition, conventional implicature, evidentials, expressives, NPI licensing, and co-speech gestures (e.g., Faller 2002; 2019; Horn 2002; Potts 2005; Simons et al. 2010; Lee 2011; Tonhauser et al. 2018; Esipova 2019; Korotkova 2020; Barnes & Ebert 2023). The concept is generally understood as distinguishing the main point of an utterance (the at-issue content) from propositions conveying background information (not-at-issue content). However, there is no consensus on how at-issueness is defined, and different diagnostics reflect different assumptions about properties of at-issue content (e.g., Tonhauser 2012; Snider 2017a; b; 2018; Tonhauser et al. 2018; Koev 2018; Faller 2019; Esipova 2019; Korotkova 2020).

Two prominent definitions of at-issueness are given in (1) and (2). The first one assumes that discourse is structured around *Questions Under Discussion* (QUDs; Roberts 2012; Ginzburg 1996) and defines at-issue content as utterance content intended to address a QUD (Amaral et al. 2007; Simons et al. 2010):

(1) QUD at-issueness: (based on Simons et al. 2010, p. 323)
A content m is at-issue in a context c iff
a. m is relevant[2] to the QUD in c, and
b. the speaker has a recognizable intention to address the QUD via m.

[1] Potts (2015) credits the term to Bill Ladusaw, "who began using it informally in his UCSC undergraduate classes in 1985" (p. 2).

[2] m may be a proposition or a question denotation, and relevance to the QUD can be defined in terms of contextual entailment of a partial or complete answer (Roberts 2012; Simons et al. 2010), or based on inferences about the probability of QUD-answers (Büring 2003; Beaver & Clark 2008).

The second one,[3] given in (2), is grounded in understanding an assertion as a proposal to update the common ground (Stalnaker 1978; Clark & Schaefer 1989; Ginzburg 1995; Farkas & Bruce 2010). Here, the at-issue content of an assertion is defined as the content that is proposed for addition to the common ground (Murray 2010; 2014; AnderBois et al. 2010; 2015, Koev 2013), while not-at-issue content is already entailed by the common ground (e.g., Stalnaker 1973; 2002) or added to the common ground without being proposed (as in Murray 2010; 2014; AnderBois et al. 2010; 2015).

(2) Proposal at-issueness: (Koev 2013, pp. 50–51)
A proposition *p* is at-issue in a context *c* iff
a. *p* is a proposal in *c* and
b. *p* has not been accepted or rejected in *c*.

Diagnostics of at-issueness operationalize the definitions in (1) and (2) via properties of at-issue content. Three properties, summarized in Tonhauser 2012, are given in (3):

(3) Properties of at-issue content (Tonhauser 2012)
a. At-issue content addresses the question under discussion (e.g., Amaral et al. 2007; Simons et al. 2010).
b. The at-issue content of questions determines the relevant set of alternatives.
c. At-issue content can be directly assented or dissented with (e.g., Shanon 1976; Groenendijk & Stokhof 1984; Faller 2002; 2006; Murray 2010).

While Tonhauser 2012 assumed that diagnostics based on all three properties in (3) measure QUD-based at-issueness in (1), other works assume that diagnostics based on properties (3a) and (3b) measure QUD-based at-issueness, whereas diagnostics based on property (3c) are either assumed to measure proposal-based at-issueness in (2) (see, e.g., Koev 2018) or rather anaphoric availability (see, e.g., Snider 2017b; Korotkova 2020).

Four commonly used diagnostics that build on the properties in (3) are illustrated in (4)–(7) for sentence-medial non-restrictive relative clauses (NRRCs), which are typically taken to contribute not-at-issue content (Potts 2005). The QUD diagnostic in (4) is based on property (3a), the 'asking whether' diagnostic in (5) is based on property (3b), and the direct-dissent and 'yes, but' diagnostics in (6) and (7) are based on property (3c).

(4) QUD diagnostic
Nora: *What did Greg buy?*
Leo: *Greg, who bought a new car, is envied by his neighbor.*
Question to participants: How well does Leo's response fit Nora's question?

(5) 'asking whether' diagnostic
Nora: *Is Greg, who bought a new car, envied by his neighbor?*
Question to participants: Is Nora asking whether Greg bought a new car?

(6) Direct-dissent diagnostic
Nora: *Greg, who bought a new car, is envied by his neighbor.*
Leo: *No, that's not true, he didn't buy a new car.*
Question to participants: How natural is Leo's rejection of Nora's utterance?

(7) 'yes, but' diagnostic
Nora: *Greg, who bought a new car, is envied by his neighbor.*

[3] Further definitions proposed in the literature include those based on restrictive vs. non-restrictive modification (Esipova 2019; 2021), and coherence-based discourse structure (Hunter & Asher 2016; Jasinskaja 2016; Koev 2018).

Leo: *Yes, but he didn't buy a new car. /*
Yes, and he didn't buy a new car. /
No, he didn't buy a new car.

Task for participants: Choose the response that sounds best.

The QUD-diagnostic in (4) targets property (3a) by testing whether the target content can felicitously address an established QUD. It presents the content in a response to a question that makes the target content relevant and elicits ratings of question-answer fit (for applications, see, e.g., Amaral et al. 2007; Lee 2011; Tonhauser 2012; Chen 2024). Responses are expected to indicate a good fit when the content can be construed as at-issue and a bad fit otherwise. Accordingly, responses are expected to indicate a bad fit for the NRRC in (4).

The 'asking whether' diagnostic in (5) targets property (3b) by probing whether the target content determines the alternatives of a polar interrogative (for applications, see, e.g., Tonhauser et al. 2018; Solstad & Bott 2024; Degen & Tonhauser 2025). It presents the target content in a polar interrogative, and elicits judgments about whether the speaker's question is about that content. These judgments are expected to indicate that the speaker is asking about the target content for at-issue content and that the speaker is not asking about the target content otherwise. Accordingly, participants are expected to judge that the speaker is not asking about the NRRC content in (5).

The direct-dissent diagnostic in (6) and the 'yes, but' diagnostic in (7) target property (3c) (for applications, see, e.g., Faller 2002; 2006; Papafragou 2006; Amaral et al. 2007; Murray 2010; 2014; AnderBois et al. 2010; 2015; Tonhauser 2012; Syrett & Koev 2015 and Xue & Onea 2011; Cummins et al. 2013; Destruel et al. 2015). Both present the target content in a declarative sentence and assume that only at-issue content can be directly assented or dissented with. The direct-dissent diagnostic elicits naturalness ratings for a response directly dissenting with the target content; the response is expected to indicate naturalness for at-issue content and unnaturalness for not-at-issue content. Accordingly, responses indicating unnaturalness are expected for the NRRC content in (6). The 'yes, but' diagnostic presents a forced choice between *'no'*, *'yes, but'*, and *'yes, and'* when denying the target content. Under the assumption that both *'yes'*-responses provide indirect denials suitable for dissenting with not-at-issue content (Xue & Onea 2011), participants are expected to prefer one of the *'yes'*-responses when denying not-at-issue content, such as the NRRC content in (7). On the other hand, participants are expected to prefer the direct denial with *'no'* when dissenting with at-issue content.[4]

At present, there is no consensus on whether the two definitions of at-issueness in (1) and (2) define the same concept, or whether diagnostics based on the three properties in (3) measure the same concept. Some works suggest a unified view. For instance, AnderBois et al. (2015) and Faller (2019) suggested that if proposing content for the common ground is intrinsically linked to managing the QUD (e.g., Farkas & Bruce 2010), the proposal-based definition in (2) can be seen as a special case of the QUD-based definition in (1) when applied to assertions. And, as mentioned above already, Tonhauser (2012) suggested that diagnostics based on the three properties in (3) can be taken to probe a QUD-based definition of at-issueness.

On the other hand, there are works that have argued that the definitions in (1) and (2) characterize different conceptions of at-issueness or that only some of the diagnostics measure at-issueness, whereas others measure anaphoric accessibility (e.g., Snider 2017b; Koev 2018; Korotkova

[4] Although at-issue content permits direct dissent, it does not require it, and indirect dissent is also possible (see, e.g., Cummins et al. 2013 on Shanon's 1976 "Hey, wait a minute" diagnostic). Work in conversation analysis suggests that preferences for direct versus indirect dissent vary with context and content. Direct dissent is often overall dispreferred (Sacks 1987; Brown & Levinson 1978; Pomerantz 1984; Locher 2004), but the strength of this preference can vary with cultural norms (e.g., Kakava 2002) or speaker gender (e.g., Tannen 1990). In contrast, direct dissent can be preferred in adversarial legal contexts (Atkinson & Drew 1979), in disputes (Kotthoff 1993; Kangasharju 2009), or in responses to self-deprecating statements (Pomerantz 1984).

2020). These positions are supported by the observation that different diagnostics may give different results for particular content. One such content is the content of sentence-medial and -final NRRCs. A forced-choice continuation experiment by Syrett & Koev (2015) found that the content of sentence-final NRRCs is more at-issue than that of medial ones, under a version of the direct-dissent diagnostic, which targets property (3c). As illustrated in (8), participants listened to (and read) utterances of declarative sentences introducing the target content (A) and chose between two *no*-responses expressing direct dissent (which were only presented in writing), targeting the content of the main clause (B1) or of the NRRC (B2). Syrett & Koev (2015: Exp. 2) reported that participants chose to dissent with the content of sentence-medial NRRCs in only 21.1% of the trials whereas they chose to dissent with the content of sentence-final NRRCs in 35.5%. These results suggest that the content of sentence-final NRRCs is more at-issue than that of medial ones.

(8) Syrett & Koev 2015, p. 541

a. Sentence-medial NRRC

A: *My friend Sophie, who performed a piece by Mozart, is a classical violinist.*

B1: *No, she's not.* (target: main clause)

B2: *No, she didn't.* (target: NRRC)

b. Sentence-final NRRC

A: *The symphony hired my friend Sophie, who performed a piece by Mozart.*

B1: *No, they didn't.* (target: main clause)

B2: *No, she didn't.* (target: NRRC)

Snider (2017b) argued that this positional effect does not emerge with diagnostics based on properties (3a) or (3b). For instance, as illustrated in (9) for the QUD diagnostic, which is based on property (3a), neither the content of the sentence-medial NRRC in B1 nor the content of the sentence-final NRRC in B2 can address A's question. This observation suggests that both contents are not-at-issue according to the QUD diagnostic, contrasting with the results of Syrett & Koev 2015: Exp. 2. Snider (2017b) proposed that diagnostics based on properties (3a) and (3b) measure at-issueness, whereas diagnostics based on property (3c) measure anaphoric accessibility (see also Snider 2017a; 2018; Korotkova 2020).

(9) Snider 2017b, p. 13

A: *Who is Margaret's cousin?*

B1:#*Pauline, who is Margaret's cousin, was interviewed by Food Network.*

B2:#*Food Network interviewed Pauline, who is Margaret's cousin.*

Similarly, Koev (2018) pointed out that diagnostics corresponding to QUD-based and proposal-based at-issueness may yield different results. For instance, whereas the content of a slifting parenthetical like *her fiancé claimed* can be directly dissented with, as illustrated in (10a), the content of the slifting parenthetical *she announced* in (10b) cannot address a salient QUD. These data again suggest that diagnostics based on property (3a) do not always give the same results as diagnostics based on property (3c). Koev 2018 proposed that the QUD-based definition of at-issueness and the proposal-based one "appear to be somewhat independent notions" (p.12).

(10) Koev 2018, p. 11

a. A: *Ellen is a passionate cook, her fiancé claimed.*

B: *No, he didn't.*

b. Q: *What did she do next?*

A: #*Her husband was a real sweetheart, she announced.*

A′: *She announced that her husband was a real sweetheart.*

Finally, a mixed picture emerged in Tonhauser et al. 2018. This work compared the 'asking whether' diagnostic in (5), based on property (3b), with a diagnostic referred to as the 'are you sure?' diagnostic, based on property (3c). On this latter diagnostic, participants read dialogues, like the one in (11) with a sentence-medial NRRC, and provide ratings about whether the final response (by Fred) addresses the *Are you sure?*-question (by Carla); this diagnostic is similar to the 'really' diagnostic of Shanon 1976. The diagnostic was taken in Tonhauser et al. 2018: 519 to be based on property (3c) because it tests whether the target content (in (11), that Mike visited Alcatraz) can be directly challenged by the *Are you sure?*-question. For the NRRC in (11), the diagnostic is expected to yield responses that indicate that the response is not addressing the question.

(11) Tonhauser et al. 2018, p. 535
Fred: *Mike, who visited Alcatraz, is a history fan.*
Carla: *Are you sure?*
Fred: *Yes, I am sure that Mike visited Alcatraz.*
Question to participants: Did Fred answer Carla's question?

On the basis of two experiments that investigated the at-issueness of contents associated with 19 English expressions, including sentence-medial NRRCs and the complements of clause-embedding predicates, like *annoyed, discover* and *know*, Tonhauser et al. (2018) found both convergence and divergence in their comparison of the results of the two diagnostics (p. 526ff.). On the one hand, they observed that mean ratings were correlated quite strongly across the two diagnostics (r=.62), suggesting that "the two ways of operationalizing at-issueness measure the same underlying concept" (p. 526.). On the other hand, the 'asking whether' diagnostic yielded lower overall ratings and smaller differentiation between contents than the 'are you sure?' diagnostic. Furthermore, for particular pairs of contents, the two diagnostics gave different results. For instance, the prejacent of *stop* was the most at-issue content on the 'asking whether' diagnostic in their Exp. 1a, but not on the 'are you sure?' diagnostic in Exp. 2a. In sum, the results of Tonhauser et al.'s 2018 comparison suggest that diagnostics based on properties (3b) and (3c) might measure the same underlying concept of at-issueness even though the diagnostics do not yield fully identical results.

In sum, it is an open question whether different diagnostics of at-issueness yield consistent results and, consequently, also whether different diagnostics measure the same underlying concept. This paper contributes to addressing these two research questions, which are summarized in (12):

(12) Research questions
  a. Do different diagnostics of at-issueness yield consistent results?
  b. Do different diagnostics of at-issueness measure the same underlying concept?

The paper proceeds as follows. §2 reports the results of four experiments designed to compare the four diagnostics in (4) to (7). The results of these experiments (Exps. 1–4) suggest that diagnostics are not interchangeable. Furthermore, the results suggest that the 'asking whether' diagnostic (Exp. 2) shows greater differentiation between some of the contents investigated than the other three diagnostics. To explore whether this greater differentiation is due to the fact that contents under the 'asking whether' diagnostic are embedded in polar interrogatives or due to the response task, §3 presents the results of two further experiments (Exps. 5–6) that compare the 'asking whether' diagnostic with a diagnostic in which the contents are also realized in polar interrogatives but where the response task taps into property (3c). The ratings elicited under the two diagnostics are highly correlated, which suggests that the high differentiation in Exp. 2 was driven by the polar interrogative embedding. In §4, we present methodological implications of this research, and also address the research questions in (12). Here, we caution against concluding that differences between the results of two diagnostics means that they measure distinct underlying concepts (cf., Snider 2017b; Koev 2018). The paper concludes in §5.

2 Experiments 1-4

To compare the results of at-issueness diagnostics,[5] we conducted four experiments that each measured at-issueness with a different diagnostic, namely the QUD diagnostic (Exp. 1, (4)), the 'asking whether' diagnostic (Exp. 2, (5)), the direct-dissent diagnostic (Exp. 3, (6)) and the 'yes, but' diagnostic (Exp. 4, (7)).[6] Each experiment tested contents associated with the same set of seven of expressions in (13): the contents of sentence-medial and sentence-final NRRCs (13a–b), as well as the contents of the clausal complements of *be right, discover, confess, confirm, discover,* and *know* illustrated in (13c) to (13g).

(13)
- a. Content of sentence-medial NRRC
  Lucy, who broke the plate, apologized. $\rightsquigarrow$ Lucy broke the plate
- b. Content of sentence-final NRRC
  The police found Jack, who saw the murder. $\rightsquigarrow$ Jack saw the murder
- c. Content of the clausal complement of *know*
  Ann knows that Raul cheated on his wife. $\rightsquigarrow$ Raul cheated on his wife
- d. Content of the clausal complement of *discover*
  Mary discovered that Denny ate the last cupcake. $\rightsquigarrow$ Denny ate the last cupcake
- e. Content of the clausal complement of *be right*
  Tom is right that Ann stole the money. $\rightsquigarrow$ Ann stole the money
- f. Content of the clausal complement of *confirm*
  Harry confirmed that Greg bought a new car. $\rightsquigarrow$ Greg bought a new car
- g. Content of the clausal complement of *confess*
  Lucy confessed that Dustin lost his key. $\rightsquigarrow$ Dustin lost his keys

These seven contents were chosen because prior literature observed differences in at-issueness between two or more of these contents using a particular diagnostic for at-issueness. As discussed in §1, Syrett & Koev 2015 observed differences between sentence-medial and -final NRRCs using a variant of the direct-dissent diagnostic. The five clause-embedding predicates were chosen based on the results of Exp. 1 in Degen & Tonhauser 2025, which implemented the 'asking whether' diagnostic. The results of this experiment, which are shown in Fig. 1, suggested great differentiation in the at-issueness of the contents of the clausal complements of *know* (which had among the lowest at-issue ratings), *discover, confess, confirm* and *be right* (which had the highest at-issue rating).

Following prior work (e.g., Tonhauser et al. 2018), we describe higher/lower mean ratings by saying that content is more/less at-issue under a given diagnostic. However, we remain agnostic about whether at-issueness is an underlyingly binary or a gradient property (for discussion, see, e.g., Tonhauser et al. 2018; Barnes & Ebert 2023). On a gradient view, gradient mean ratings may be taken to reflect gradient at-issueness. On a categorical view, gradient mean ratings could be taken to reflect uncertainty about what the relevant QUD or proposal is, thus indicating the frequency or ease with which a particular QUD is attributed to the utterance or whether with which the content is taken to be a proposal.

[5] We use the term "at-issueness diagnostic" descriptively throughout the paper, even though our findings may ultimately suggest that some diagnostic tracks a different theoretical construct. We return to this issue in the general discussion.

[6] The experiments, data and R code for generating the figures and analyses of the experiments reported in this paper are available at https://anonymous.4open.science/r/at-issueness-diagnostics-232C/.

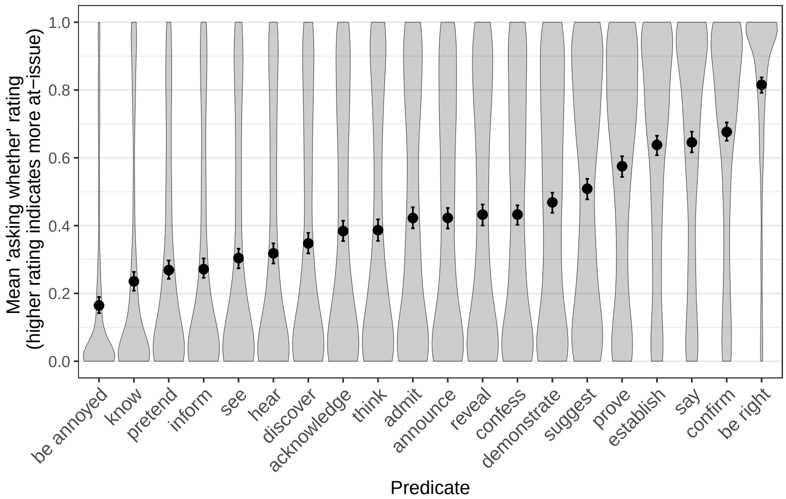


Figure 1: Mean 'asking whether' ratings for the contents of the clausal complements of 20 clause-embedding predicates, based on Degen & Tonhauser 2025: Exp. 1. Error bars indicate 95% bootstrapped confidence intervals. Violin plots indicate the kernel probability density of individual participants' ratings.

2.1 Methods

2.1.1 Participants

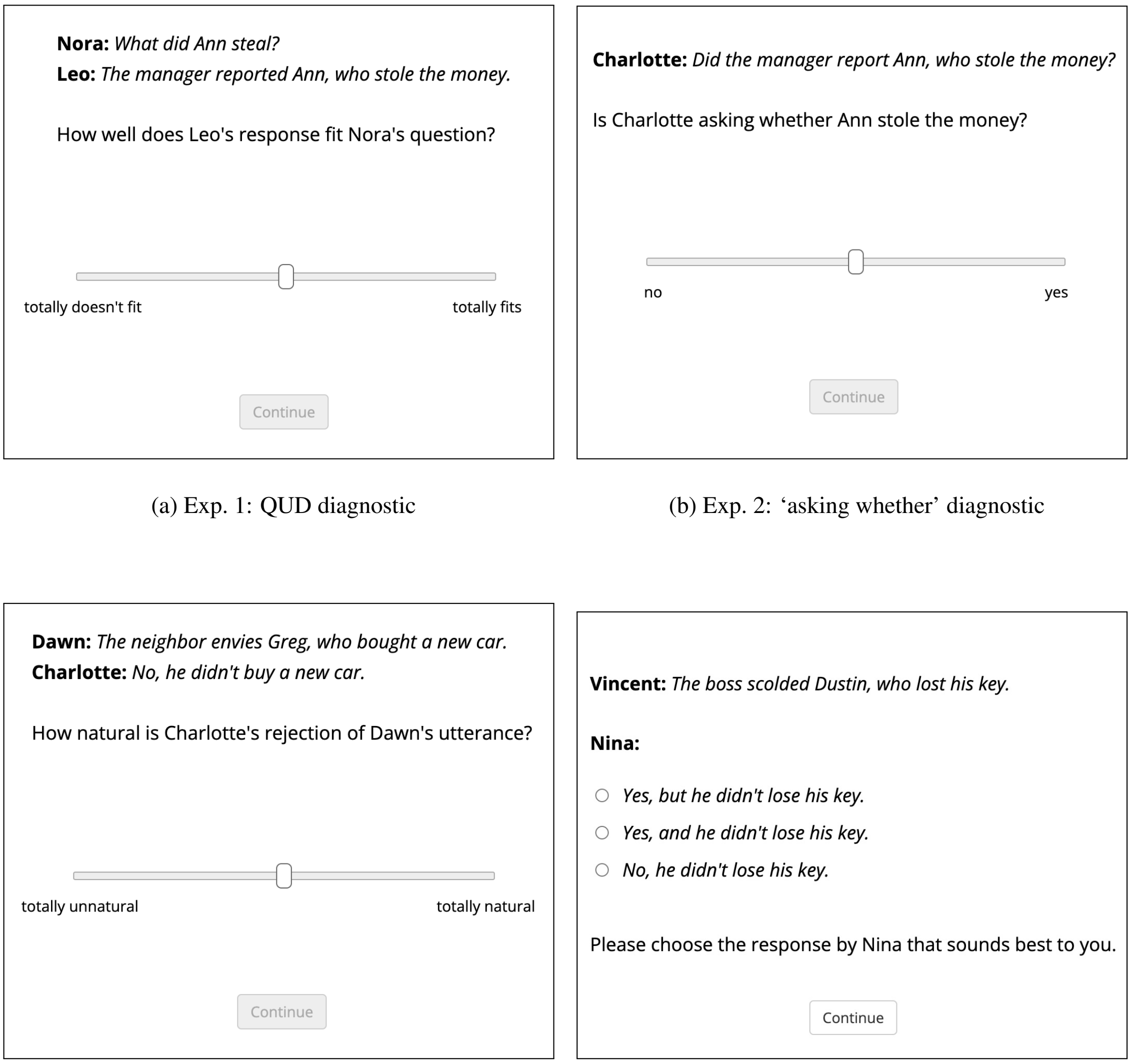
For each of the four experiments, we recruited 80 unique participants on Prolific. These participants had registered on the platform as living in the USA and as having English as their primary language. They had at least 50 previous submissions and an approval rate of at least 97%. Table 1 shows the age and gender distributions of the recruited participants.

	recruited	ages (mean age)	f/m/nb/dnd
Exp. 1 (QUD)	80	18-81 (43.8)	42/37/0/1
Exp. 2 (asking whether)	80	20-74 (38.5)	48/30/1s/1
Exp. 3 (direct dissent)	80	18-77 (39.1)	50/28/1/1
Exp. 4 (yes, but)	80	19-67 (38.0)	48/30/2/0

Table 1: Information about the participants recruited in Exps. 1-4 (f = female, m = male, nb = nonbinary, dnd = did not disclose).

2.1.2 Materials and procedure

Fig. 2 provides sample trials from each of the four experiments. As shown, the response options differed by diagnostic. In Exp. 1 (QUD diagnostic, panel (a)), participants rated how well the response fit the question on a slider marked 'totally doesn't fit' on one end (coded 0) and 'totally fits' on the other end (coded as 1). In Exp. 2 ('asking whether' diagnostic, panel (b)), participants judged whether the question was about the target content, using a slider marked 'no' on one end (coded as 0) and 'yes' on the other (coded as 1). In Exp. 3 (direct-dissent diagnostic, panel (c)), participants rated the naturalness of the direct dissent on a slider marked 'totally unnatural' (coded as 0) on one end and 'totally natural' on the other (coded as 1). Finally, in Exp. 4 ('yes, but' diagnostic, panel (d)), participants chose the response that sounded best; the two indirect dissents were coded as 0 and the direct one as 1. Across the four experiments, the responses were coded so that 1 meant that the content to be diagnosed was rated as at-issue and 0 as not-at-issue.



(a) Exp. 1: QUD diagnostic

(b) Exp. 2: 'asking whether' diagnostic

(c) Exp. 3: direct-dissent diagnostic

(d) Exp. 4: 'yes, but' diagnostic

Figure 2: Sample trials in (a) Exp. 1, (b) Exp. 2, (c) Exp. 3, and (d) Exp. 4.

Each of the seven contents in (13) was instantiated by the seven items in (14) in each of the four experiments.

(14)
a. Jack saw the murder.
b. Raul cheated on his wife.
c. Ann stole the money.
d. Danny ate the last cupcake.
e. Lucy broke the plate.
f. Dustin lost his key.
g. Greg bought a new car.

Each experiment included two control stimuli serving as attention checks: one was expected to receive a response at one end of the slider (Exps. 1-3) or a 'no' response (Exp. 4); the other control stimulus was expected to receive a response at the other end of the slider (Exps. 1-3) or a 'yes' response (Exp. 4). See Supplement A for the control stimuli used in Exps. 1-4.

In all four experiments, each participant's set of items was generated by randomly realizing each of the seven contents in (13) using a unique item in (14). Participants completed a total of 9 trials, namely 7 target trials and the same 2 control trials. Trial order was randomized for each participant.

After completing the experiment, participants filled out a short optional demographic survey. To encourage truthful responses, participants were told that they would be paid no matter what answers they gave in the survey.

2.1.3 Data exclusion

We excluded the data of participants that were not self-reported native speakers of American English and of participants whose responses to one of the two control trials was more than 2 sd away from the group mean (Exps. 1–3) or whose responses to one of the two control trials was incorrect (Exp. 4). Table 2 shows how many participants were excluded in each experiment, demographic information for the remaining participants, and the number of data points that entered into the analyses.

	exclusion criterion		**remaining participants**		data
	language	fillers	ages (mean age)	f/m/nb/dnd	points
Exp. 1 (QUD)	1	10	18-81 (41.1)	36/32/0/1	621
Exp. 2 (asking whether)	2	4	22-74 (38.7)	45/27/1/1	666
Exp. 3 (direct dissent)	2	7	18-77 (39.5)	44/25/1/1	639
Exp. 4 (yes, but)	4	4	19-67 (38.5)	43/27/2/0	648

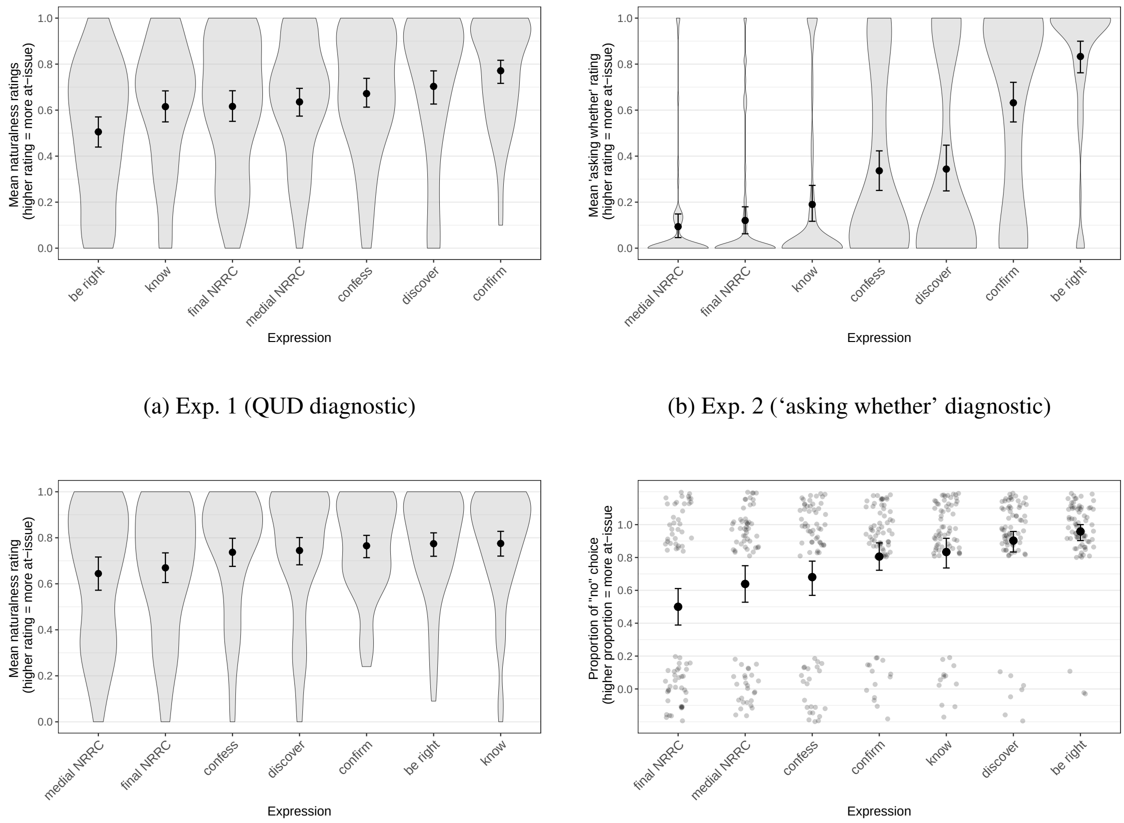
Table 2: Information from Exps. 1-4 about the number of participants whose data was excluded based on their self-declared language (variety) and the fillers, about the remaining participants, and about the number of data points that entered into the analysis.

2.2 Results

The four panels of Fig. 3 plot the results of the four experiments by the expressions that are associated with the seven target contents. As shown, the diagnostics differ in the overall range of mean responses across the seven contents (given as the difference between the largest and smallest by-content means). The range is largest in Exp. 2 ('asking whether'), at .74 (from .1 to .83) and smallest in Exp. 3 (direct dissent), at .13 (.64 to .78). The results of Exp. 1 (QUD diagnostic), with a range of .27 (.51 to .77) and Exp. 4 ('yes, but'), with a range of .46 (.5 to .96), fall in between.

The results of the experiments also differ in whether they replicated distinctions between contents reported in prior work. First, regarding the content of sentence-medial and -final NRRCs, none of the results of Exps. 1–4 provide empirical support for the content of sentence-final NRRCs being more at-issue than that of sentence-medial ones, in contrast to the results of Syrett & Koev (2015: Exp. 2). In fact, the results of Exp. 4 even suggest that the content of sentence-medial NRRCs is more at-issue than that of sentence-final ones. Overall, then, these results suggest that different diagnostics for at-issueness are not interchangeable when it comes to sentence-medial vs. -final NRRCs.

For clause-embedding predicates, Exp. 2 ('asking whether') largely replicated the pattern from Degen & Tonhauser (2025): the content of the complement of *know* was rated as less at-issue than that of *discover*, followed by *confess*, then *confirm*, and finally *be right*. The only difference between the results of Degen & Tonhauser (2025) and Exp. 2 is that there is no support in Exp. 2 that the content of the complement of *confess* and *discover* differ in at-issueness. The results of Exps. 1, 3 and 4 supported fewer differences between the contents of the five clausal complements.



(a) Exp. 1 (QUD diagnostic)

(b) Exp. 2 ('asking whether' diagnostic)

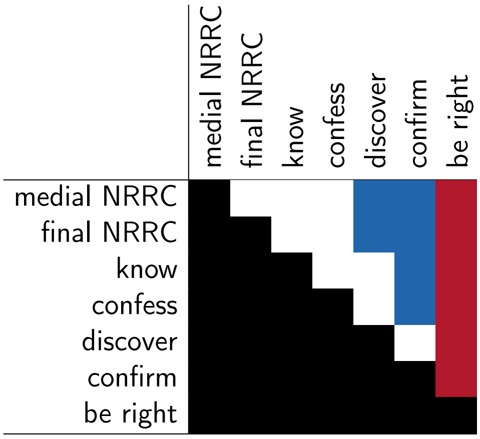
(c) Exp. 3 (direct-dissent diagnostic)

(d) Exp. 4 ('yes, but' diagnostic)

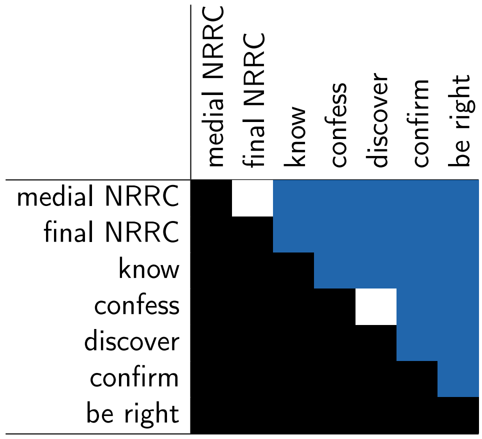
Figure 3: Results of Exps. 1–4. Panels (a)–(c) show the mean responses by expression for (a) Exp. 1 (QUD diagnostic), (b) Exp. 2 ('asking whether' diagnostic), and (c) Exp. 3 (direct-dissent diagnostic); panel (d) shows the proportion of 'no' choices by expression in Exp. 4 ('yes, but' diagnostic). Error bars indicate 95% bootstrapped confidence intervals. Violin plots in panels (a)-(c) show the kernel probability density of individual participants' ratings. Gray dots in panel (d) represent individual participant responses ('no' vs. 'yes', jittered vertically and horizontally for legibility).

In Exp. 1 (QUD diagnostic), the contents of the complements of four of the five predicates showed little differentiation, with the exception of that of *confirm*. Unexpectedly, the content of the complement of *be right* was rated least at-issue, in contrast to the results of Degen & Tonhauser 2025. The results of Exp. 3 show no clear differences among the contents of the complements of the five clause-embedding predicates. Finally, the results of Exp. 4 showed little differentiation between *confirm, know* and *discover*; the highest proportion of 'no' answers was observed for *be right*, the lowest for *confess*. In sum, the results of the four experiments for the contents of the complements of the five clause-embedding predicates suggest that the diagnostics they implement are not interchangeable.

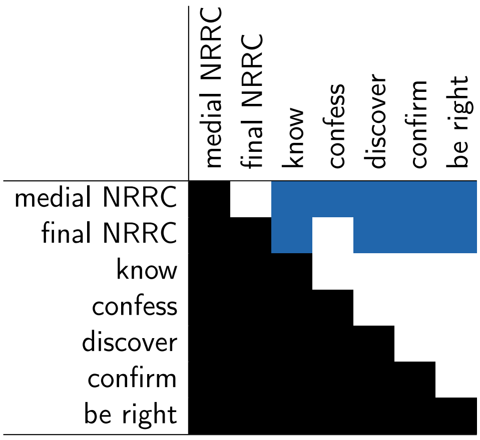
These observations are supported by post-hoc pairwise comparisons of the estimated means or proportions for each content, using the `emmeans` package (Lenth 2023) in R (R Core Team 2016), illustrated in Fig. 4. These comparisons were based on mixed-effects beta regression models (Exps. 1–3) and a mixed-effects logistic regression model (Exp. 4), all fit using the `brms` package



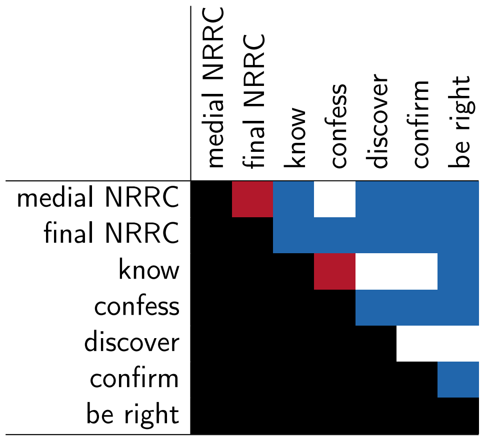
(a) Exp. 1 (QUD diagnostic)



(b) Exp. 2 (‘asking whether’ diagnostic)



(c) Exp. 3 (direct-dissent diagnostic)



(d) Exp. 4 (‘yes, but’ diagnostic)

Figure 4: Pairwise differences between expressions, ordered from top to bottom and left to right by increasing mean in Exp. 2 (‘asking whether’ diagnostic). A white cell means that the 95% HDI of the pair of the row expression and the column expression includes 0 (i.e., the two contents do not differ), a red cell means that the 95% HDI does not include 0 and that the coefficient is positive (the row expression received a higher rating than the column expression), and a blue cell means that the 95% HDI does not include 0 and the coefficient is negative (the row expression received a lower rating than the column expression).

(Bürkner 2017) using weakly informative priors. The models predicted the ratings[7] from a fixed effect of expression (treatment-coded, with *be right* as reference level), with random intercepts for participants and items. The output of the pairwise comparison were 95% highest density intervals (HDIs) of estimated marginal mean differences between each of the expressions. We assume that two contents differ if their HDI does not include 0.[8]

Consistent with the patterns described above, Fig. 4 shows that the only positional difference for NRRCs is observed in Exp. 4 (‘yes, but’), where sentence-medial ones are rated more at-issue

[7] To model the ratings in Exps. 1–3 using a beta regression, the ratings were first transformed from the interval [0,1] to the interval (0,1) using the method proposed in Smithson & Verkuilen 2006.

[8] The full model outputs are available in the folder `results+analysis/exps1-4/` in the repository available at https://anonymous.4open.science/r/at-issueness-diagnostics-232C/.

than sentence-final ones, opposite to the finding in Syrett & Koev (2015). For the contents of the complements of the clause-embedding predicates, recall that the five predicates were chosen because Degen & Tonhauser 2025: Exp. 1, which used the 'asking whether' diagnostic, found differences between all of them: the content of the complement of *be right* was most at-issue, followed by those of *confirm*, *confess*, and *discover*, with the content of the complement of *know* least at-issue; see Fig. 1. The results of Exp. 2 ('asking whether') came closest to reproducing this pattern, with 9 out of 10 possible pairwise differences supported, as shown in panel (b) of Fig. 4. In contrast, the results of Exps. 1 (QUD) and 4 ('yes, but') only supported 2 and 5 of the 10 possible pairwise differences, and the results of Exp. 3 (direct-dissent) supported none.

In sum, the four diagnostics implemented in Exps. 1–4 do not yield consistent results. None of them replicated Syrett & Koev's 2015 result for sentence-medial vs. -final NRRCs, and they yielded inconsistent results for the contents of the complements of the five clause-embedding predicates investigated.

2.3 Discussion

Exps. 1–4 were designed to compare the four at-issueness diagnostics in (4) to (7), which have been widely used in the literature. The results of Exps. 1–4 suggest that diagnostics for at-issueness are not readily interchangeable. This section discusses these results.

2.3.1 At-issueness differences between sentence-medial vs. -final NRRCs

We begin with the question of why the results of none of our Exps. 1–4 replicated the difference between sentence-medial and -final NRRCs observed in Syrett & Koev 2015: Exp. 2.

For Exp. 2 (asking whether'), we hypothesize that the absence of a positional effect may be due to an interaction between the diagnostic and the illocutionary properties of NRRCs, namely that they contribute a speech act that is separate from the speech act conveyed by their host sentence (see, e.g., Potts 2005; Koev 2013; Syrett & Koev 2015). Under this view, the sample stimuli of Exp. 2 shown in (15) convey two distinct speech acts: the content of the NRRC is an assertion and the main clause content is an (information-seeking) question.

(15) Sentence-medial NRRC: *Does Greg, who bought a new car, have a driver's license?*
Sentence-final NRRC: *Does the neighbor envy Greg, who bought a new car?*
Question for participants: Is the speaker asking whether Greg bought a car?

If the NRRC contributes a separate assertion rather than being part of the question, participants would not be expected to judge the speaker as asking about its content, regardless of whether it is contributed sentence-medially or -finally. This hypothesis is consistent with the very low mean ratings for both medial and sentence-final NRRCs in Exp. 2.

In the other three experiments, target contents were presented in declarative main clauses, which contribute assertions. In discussing their findings, Syrett & Koev (2015) proposed that NRRC content competes with that of the main clause for at-issue status. If so, NRRC content should, in principle, be able to compete with the content of the main clause for at-issue status in Exps. 1, 3 and 4. However, as pointed out above, the results of these experiments do not support the positional difference observed in Syrett & Koev 2015 either. This is true even for the dissent-based diagnostics implemented in Exps. 3 (direct dissent) and Exp. 4 ('yes, but'), which most closely resemble the diagnostic used in Exp. 2 of Syrett & Koev (2015), as they all rely on property (3c) of at-issue content. There are several differences between the experiments that may have contributed to this divergence in results. One of them are the the items used. Another one is that the target sentences in Syrett & Koev 2015: Exp. 2 were not only presented in writing to the participants, as in our

Exps. 1–4, but also in sound. Another difference concerns the response task. Although Syrett & Koev 2015: Exp. 2 and our Exps. 3 and 4 used dissent-based diagnostics with target content presented in declarative assertions, the response tasks of the three experiments differed. Specifically, Syrett & Koev 2015: Exp. 2 forced participants to choose between a dissent with the content of the NRRC and that of the main clause, Exp. 3 elicited naturalness ratings on question/answer pairs, and Exp. 4 used a force choice task, but the response options differed from those in Syrett & Koev 2015 in that participants had to choose between different forms of dissent with the content of the NRRC. A natural follow-up would be to investigate whether the diagnostic as implemented in Syrett & Koev 2015: Exp. 2 applied to our stimuli replicates the distinction between sentence-medial and -final NRRCs that they observed, and whether presenting spoken vs. written stimuli is critical. Importantly, we argue in §4 that it would be premature to conclude from these differences in the results that these diagnostics do not measure the same underlying concept.

2.3.2 An interaction between the QUD diagnostic and the meaning of *be right*

For the clause-embedding predicates, one of the most salient differences observed between the results of Exps. 1–4 concerns the content of the complement of *be right*, which received the lowest ratings in Exp. 1 (QUD diagnostic) and relatively high ratings in Exps. 2–4 as well as Degen & Tonhauser 2025. We hypothesize that the low mean rating for the content of the complement of *be right* in Exp. 1 is due to an interaction between the QUD diagnostic and the meaning of *be right*.

To illustrate, consider the examples in (16), which present a sample stimulus from each of the four experiments with *be right*:

(16) Sample stimuli from the four experiments with *be right*

a. Exp. 1 (QUD diagnostic)
Nora: *What did Lucy break?*
Leo: *Danny is right that she broke the plate.*

b. Exp. 2 (‘asking whether’ diagnostic)
Nora: *Is Danny right that Lucy broke the plate?*

c. Exp. 3 (‘direct dissent’ diagnostic)
Nora: *Danny is right that Lucy broke the plate.*
Leo: *No, she didn’t break the plate.*

d. Exp. 4 (‘yes, but’ diagnostic)
Nora: *Danny is right that Lucy broke the plate.*
Leo: *Yes, but she didn’t break the plate.*
Yes, and she didn’t break the plate.
No, she didn’t break the plate.

We hypothesize that the low ratings in Exp. 1 arise because utterances which include *be right* in a veridical environment, like Leo’s utterance in (16a), convey that the attitude holder (here, Danny) has publicly committed to the content of the clausal complement (here, that Lucy broke the plate),[9] This commitment of Danny has been made explicit in the variant of (16a) in (17):

(17) **Danny:** *Lucy broke the plate.*
Nora: *What did Lucy break?*
Leo: *Danny is right that she broke the plate.*

[9] Abusch (2002; 2010) characterizes *be right* as conveying a projective doxastic inference, namely that the subject believes the embedded proposition. Following a suggestion in Schlenker 2010 (p. 135), Anand & Hacquard (2014: p. 73) argue that the content of this projective inference instead reports an assertive speech act, conveying that the subject has committed to the embedded content by means of a prior discourse move.

We can understand the ill-formedness of Leo's utterance in (16a)/(17) within a QUD-based framework of discourse structure (see, e.g., Ginzburg 1996; Roberts 2012; Büring 2003). In this framework, Danny's utterance in (17) is assumed to address the question whether Lucy broke the plate.[10] The inference associated with Leo's utterance, that Danny has previously committed to Lucy breaking the plate, can therefore be interpreted within a QUD-framework as conveying that the question "whether Lucy broke the plate" is on the QUD-stack of previously raised questions at the point at which Leo makes his utterance. However, this QUD is a subquestion of Nora's question, since every complete answer to the latter entails an answer to the former. Crucially, then, (16a)/(17) involves a progression from a subquestion to an explicitly given superquestion, which violates Roberts's 2012 constraint on QUD stacks (p.15), which allows only the reverse order, from superquestions to subquestions.[11] Thus, the low ratings for *be right* in Exp. 1 need not indicate that content of the clausal complement is not at-issue but may be due to an infelicitous discourse context.

In the other diagnostics (16b)–(16d), where no prior discourse is given, the ratings for *be right* are not affected by its inference about a previous discourse commitment. This difference between the results of Exp. 1, on the one hand, and Exps. 2–4, on the other, highlights that diagnostics may interact with utterance meaning independently of at-issueness, and such interactions must be taken into account when interpreting the results.

2.3.3 Clause-embedding predicates: 'asking whether' vs. the other diagnostics

We now turn to the observation that the results of Exp. 2 ('asking whether') showed greater differentiation than the other three experiments between the contents of the complements of the five clause-embedding predicates. Recall that the five predicates were chosen because Degen & Tonhauser 2025: Exp. 1, which also used the 'asking whether' diagnostic, found differences between them. Whereas the results of Exp. 2 came close to replicating this pattern, the results of the other three experiments did not.

One difference between the 'asking whether' diagnostic (Exp. 2) and the other diagnostics (Exps. 1, 3, and 4) is the clause type in which the target content is realized, namely polar interrogatives in Exp. 2 and positive declaratives in the other three. Another difference is the response task. The response task of the 'asking whether' diagnostic in Exp. 2 asked participants to give a metalinguistic judgment about what the speaker was asking about. By contrast, the response tasks of the other three diagnostics asked participants to give a naturalness rating about the fit of a response to a question (Exp. 1, QUD diagnostic) or about a rejection (Exp. 3, direct dissent diagnostic), or it asked participants to choose among three dissent options (Exp. 4, 'yes, but' diagnostic). There are, therefore, at least two hypotheses about why Exp. 2 showed greater differentiation among the contents of the complements of the five clause-embedding predicates: On the first hypothesis, the greater differentiation is due to the fact that the 'asking whether' hypothesis as implemented in Exp. 2 presents the contents in polar interrogatives rather than declaratives; on the second hypothesis, the greater differentiation is due to the the fact that the 'asking whether' hypothesis as implemented in Exp. 2 asked participants to judge what the speaker was asking about. The two experiments presented in the next section allow us to distinguish between these two hypotheses.

[10] While asserting p is typically assumed to raise the issue of whether p (see, e.g., Ginzburg & Sag 2000; Farkas & Bruce 2010), this may well be a subquestion (i.e., part of an answering strategy in the sense of Roberts 2012) of a superquestion such as "What did Lucy break?" or *Who broke the plate?*, since an answer to *whether p* entails a partial answer to either of the latter two (see definition of subquestion in Roberts 2012, p.6:16).

[11] This constraint on QUD stacks can be understood as an informativeness-constraint, which requires that discourse moves contribute towards increasing the information quantity in the context set. Accordingly, the exchange in (17) is acceptable only under an echo-question interpretation of Nora's question.

3 Experiments 5 and 6

In Exps. 5 and 6, target contents were realized in polar interrogatives: Exp. 5 implemented the 'asking whether' diagnostic as in Exp. 2, as shown in (18a), and Exp. 6 implemented a diagnostic that we will refer to as the 'affirmative response' diagnostic. Under this diagnostic, target contents are embedded in polar interrogatives, but the stimuli additionally consist of a response by a named speaker who affirms the target content. As illustrated in (18b), participants are asked to rated the naturalness of the affirmative response (see, e.g., Amaral et al. 2007; Tonhauser 2012). Such a response is natural only if the target content provides a possible answer, that is, is a question alternative (von Stechow 1991). Accordingly, participants are expected to judge a dialogue as natural when the target content is provides a question alternative and as unnatural otherwise.

(18) Diagnostics implemented in Exps. 5–6
- a. Exp. 5 ('asking whether' diagnostic)
  Nora: *Is Tom right that Lucy broke the plate?*
  Question to participants: Is Nora asking whether Lucy broke the plate?
- b. Exp. 6 ('affirmative response' diagnostic)
  Nora: *Is Tom right that Lucy broke the plate?*
  Leo: *Yes, Lucy broke the plate.*
  Question to participants: Does Leo's response to Nora sound good?

Both diagnostics implemented in Exps. 5–6 are based on property (3b), according to which the at-issue content of a polar interrogative determines the question alternatives. But the response task of the diagnostic implemented in Exp. 6 also taps into property (3c), according to which at-issue content can be directly assented or dissented with.

Exps. 5 and 6 measured the at-issueness of the contents of the complements of the 20 clause-embedding predicates (and 20 items) from Degen & Tonhauser 2022 and Degen & Tonhauser 2025, which were shown in Fig. 1 and are repeated in (19) for convenience:

(19) 20 clause-embedding predicates from Degen & Tonhauser 2025:
acknowledge, admit, announce, be annoyed, be right, confess, confirm, demonstrate, discover, establish, hear, inform, know, pretend, prove, reveal, say, see, suggest, think

Both experiments also measured the projectivity of the contents of the clausal complements. This data (but not the at-issueness data) was reported in Hofmann et al. 2024 and is ignored here.

3.1 Methods

3.1.1 Participants

We recruited 284 participants on Amazon's Mechanical Turk platform for Exp. 5 and 250 participants on Prolific for Exp. 6.[12] Participants recruited on Mechanical Turk had U.S. IP addresses and at least 99% of previous HITs approved. Participants recruited on Prolific had registered as U.S.-born native speakers of English residing in the USA and had an approval rate of at least 99%. Table 3 summarizes the age and gender distributions of the recruited participants.

3.1.2 Materials and procedure

The two panels of Fig. 5 present a sample stimulus of Exps. 5 and 6. As shown in panel (a), target stimuli in Exp. 5 ('asking whether') consisted of a polar interrogative uttered by a named speaker.

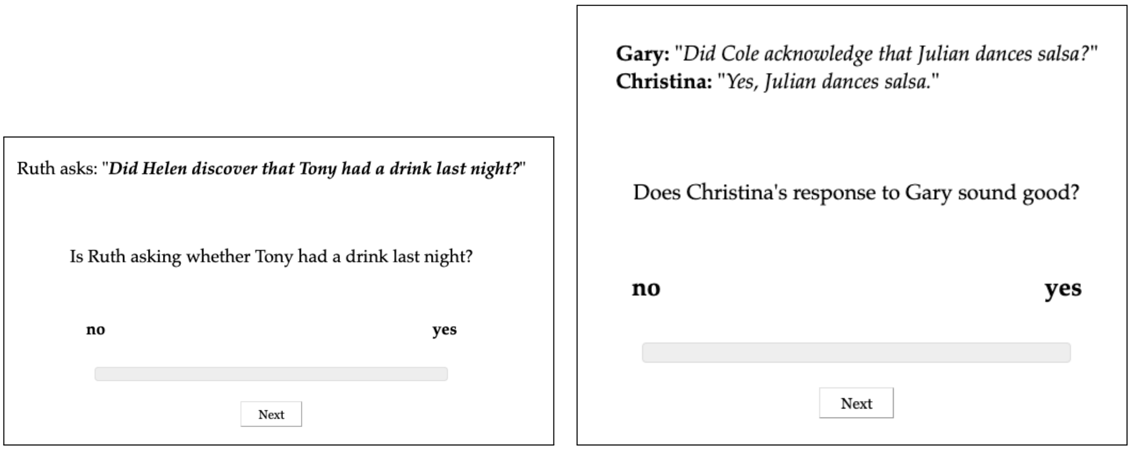
[12] Exp. 5 was run in August 2019 and Exp. 6 in August 2021.

	recruited	ages (mean age)	f/m/nb/dnd
Exp. 5 (asking whether)	284	19-74 (38.2)	–/–/–/–
Exp. 6 (direct response)	250	18-58 (25.5)	201/43/6/0

Table 3: Information about the participants recruited in Exps. 5-6 (f = female, m = male, nb = nonbinary, dnd = did not disclose; gender information was not collected in Exp. 5).

As in Exp. 2 (‘asking whether’), participants judged whether the question was about the content of the complement, using a slider marked ‘no’ on one end (coded as 0) and ‘yes’ on the other (coded as 1). As shown in panel (b), target stimuli in Exp. 6 (‘affirmative response’) consisted of a dialogue consisting of a polar interrogative and an affirmative response uttered by a named speaker. Participants rated whether the affirmative response to the first speaker sounds good, on a slider marked ‘no’ (coded as 0) on one end and ‘yes’ (coded as 1) on the other. Thus, in both experiments, responses were coded so that 1 meant that the target content was rated as at-issue and 0 as not-at-issue.

In both experiments, the polar interrogatives of the target stimuli were constructed by combining each of the 20 clause-embedding predicates in (19) with one of 20 items (see Supplement B). Both experiments included the same six control stimuli, where target contents were expected to be at-issue (see Supplement C). Participants were instructed to imagine they are at a party and that, when walking into the kitchen, they overhear somebody say something to somebody else.



(a) Exp. 5: ‘asking whether’ diagnostic

(b) Exp. 6: ‘affirmative response’ diagnostic

Figure 5: Sample trials in (a) Exp. 5 and (b) Exp. 6.

For each participant, the 20 predicates were randomly paired with the 20 items, yielding one target stimulus per predicate. Participants thus rated the at-issueness of contents in 26 stimuli in total: 20 target items and 6 controls.[13] Trial order was randomized for each participant.

After completing the experiment, participants filled out a short optional demographic survey. To encourage truthful responses, participants were told that they would be paid no matter what answers they gave in the survey.

[13] Each participant saw their set of 26 stimuli twice, once in the projection block and once in the at-issueness block. Block order was randomized. As mentioned above, we only discuss the at-issueness data here.

3.1.3 Data exclusion

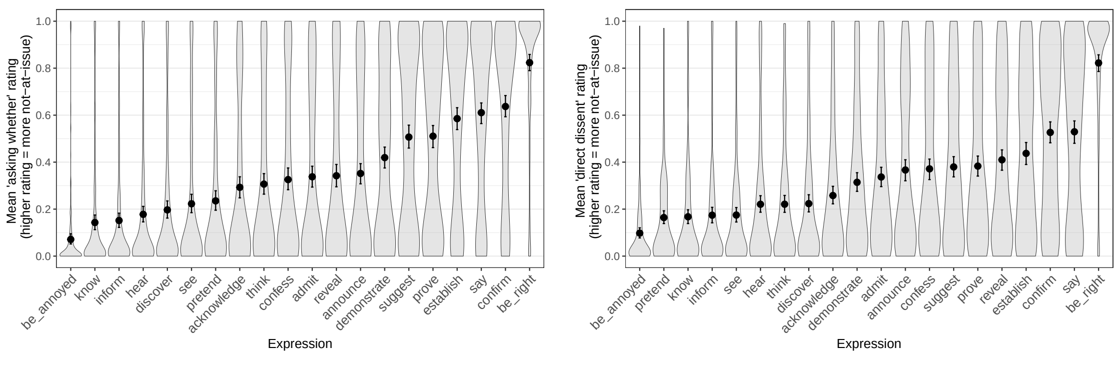
We excluded data from participants who did not self-identify as native speakers of American English, and those whose responses to one of the two control trials was more than 2 sd from the group mean. Table 4 shows the number of exclusions in each experiment, the characteristics of the remaining participants, and the number of data points that entered into the analyses.

	exclusion criterion		**remaining participants**			data
	language	controls	ages (mean age)	f/m/nb/dnd	total	points
Exp. 5	7	35	21-74 (39.2)	–/–/–/–	242	6292
Exp. 6	5	24	18-58 (24.9)	187/29/5/0	221	5746

Table 4: Information from Exps. 5-6 about the number of participants whose data was excluded based on their self-declared language and language variety ('language'), the controls, and the variance of their responses, about the remaining participants, and about the number of at-issueness data points that entered into the analysis.

3.2 Results and discussion

Fig. 6 shows the results of the two experiments by predicate: panel (a) plots mean 'asking whether' ratings in Exp. 5, and panel (b) plots mean naturalness ratings in Exp. 6 ('affirmative response'). Overall, the results of two experiments are very similar. The ranges of by-content means are comparable (Exp. 5 range: .75, .07 to .82; Exp. 6 range: .73, .09 to .82), and the relative orderings of contents by their means align closely, as reflected in a strong Spearman rank correlation ($r_s =$ .93).**JT: The y-axis labels of the two figures in Fig 6 need to be updated (higher ratings means more at-issue)**

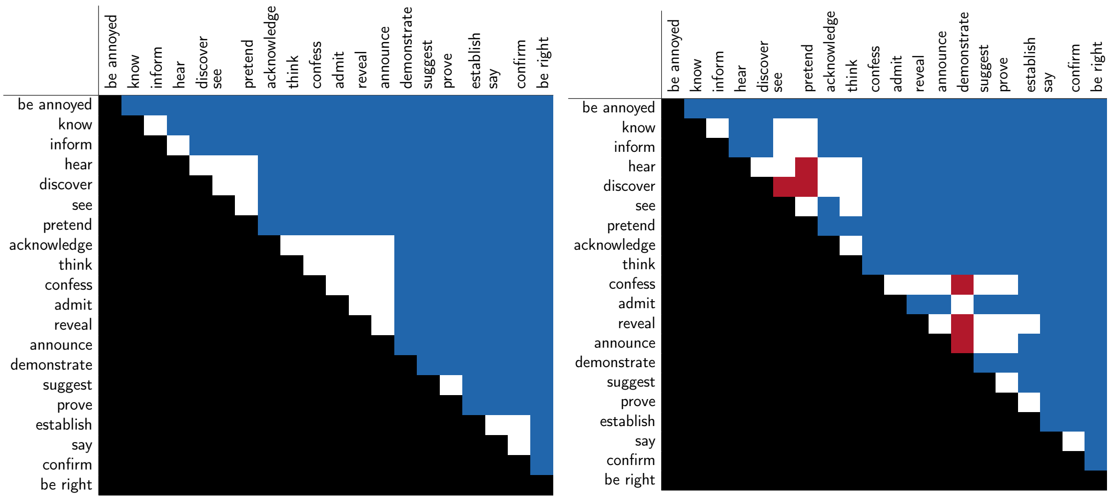


(a) Exp. 5 ('asking whether' diagnostic) (b) Exp. 6 ('affirmative response' diagnostic)

Figure 6: Mean ratings in Exps. 5–6, with predicates ordered from left-to-right by increasing mean rating. The panels show the mean ratings by expression for (a) Exp. 5 ('asking whether' diagnostic) and (b) Exp. 6 ('affirmative response' diagnostic). Error bars indicate 95% bootstrapped confidence intervals. Violin plots show the kernel probability density of individual participants' ratings.

Both experiments show fine-grained by-content differentiation, as evidenced by the results of a post-hoc pairwise comparisons of the estimated means for each content shown in Fig. 7. As in §2, we used emmeans (Lenth 2023) in R (R Core Team 2016) applied to mixed-effects beta regression models fit with brms (Bürkner 2017) using weakly informative priors. The models predicted ratings[14] from a fixed effect of content (treatment-coded, with *be right* as the reference level). As shown, the results of the two experiments yielded very similar results for the 20 contents investigated, including fine-grained by-predicate differentiation. For the five clause-embedding predicates tested in Exps. 1–4, the results of Exps. 5 and 6 reproduced the ordering found in Degen & Tonhauser (2025) and provide support for all 10 pairwise differences in the predicted directions, with increasing mean ratings between *know, discover, confess, confirm* and *be right* in that order.

There are also some differences between the experiments, concerning local reversals in the middle of the scale. For example, the content of the complement of *demonstrate* came out as more at-issue than that of *announce*, *confess*, and *reveal* in Exp. 5 ('asking whether'), but as less at-issue than the content embedded under these predicates in Exp. 6 ('affirmative response').



(a) Exp. 5 ('asking whether' diagnostic)

(b) Exp. 6 ('direct assent to polar interrogative')

Figure 7: Pairwise differences between expressions, ordered from by increasing mean in Exp. 5 ('asking whether'). White cells indicate that the 95% HDI of the difference includes 0. Red cells indicate a positive difference (the row expression received a higher rating than the column expression), and blue cells indicate a negative difference.

Overall, then, because the results of Exps. 5 and 6 are so similar, the results provide empirical support for the hypothesis on which the results of Exp. 2 ('asking whether') showed greater differentiation because the target contents are embedded in polar interrogatives. The hypothesis on which this greater differentiation is due to the response task is not supported, as the results of Exp. 6, which used a different response task than Exps. 2 and 5, are so similar to those of Exp. 5.

4 General discussion

This paper reported the results of six experiments to address two research questions, namely (12a) whether different diagnostics of at-issueness yield consistent results, and (12b) whether different diagnostics of at-issueness measure the same underlying concept. In this section, we discuss in §4.1 what the results of the six experiments suggest regarding the research question in (12a), and

[14] To model the ratings using a beta regression, the ratings were first transformed from the interval [0,1] to the interval (0,1) using the method proposed in Smithson & Verkuilen 2006.

in §4.2 what they suggest regarding the research question in (12b). Finally, in §4.3, we present methodological implications of the results.

4.1 Do different diagnostics of at-issueness yield consistent results?

We begin to address research question (12a) by taking stock. On the one hand, there are several results that suggest a negative answer to the research question: For the contents of sentence-medial and -final NRRCs, the results of Exps. 1–4 did not replicate Syrett & Koev's 2015 result, that sentence-final NRRCs are more at-issue than -medial ones. This result resonates with Snider's 2017b observation, discussed in §1. Likewise, Koev 2018 pointed out that the direct dissent and the QUD diagnostic give different results for the content of slifting parentheticals, as also discussed in §1. Finally, for the contents of the complements of the five clause-embedding predicates, only the results Exp. 2 ('asking whether') came close to replicating the results of Degen & Tonhauser 2025, which also used the 'asking whether' diagnostic. These results suggest that different diagnostics do not necessarily yield consistent results.

On the other hand, the results of Exps. 5–6 may be taken to suggest a positive answer to research question (12a): The results of these two experiments were very similar even though they differed in the diagnostic implemented and the response task; crucially, while the diagnostic implemented in Exp. 5 only tapped into property (3b), the diagnostic implemented in Exp. 6 also tapped into property (3c). The results of Tonhauser et al. 2018 also suggested that the 'asking whether' and the 'are you sure?' diagnostics (which also rely on distinct properties of at-issue content, namely property (3b) and (3c), respectively) may "measure the same underlying concept" (p. 526.), even though there also were some differences in the results of the two diagnostics. These results suggest that different diagnostics can yield consistent results.

Taken together, the results of Exps. 1–6 and those reported in prior literature suggest that the answer to research question (12a) is (an unsatisfying) "maybe": Different diagnostics may yield different results when applied to the same contents, but they also may not. An important next step in research on at-issueness is to identify when diagnostics give different results and when they do not. The research reported on in this paper already provides some useful pointers in this regard. First, even when two diagnostics rely on the same property of at-issue content, different response tasks may lead to different results (see, e.g., Syrett & Koev 2015: Exp. 2 vs. Exps. 3–4). Second, realizing target contents in polar interrogatives (as in Exps. 2, 5–6) may lead to greater differentiation between some contents (like the contents of the complements of clause-embedding predicates) than realizing target contents in positive declaratives (as in Exps. 1, 3–4). Third, when target contents are realized in polar interrogatives, it may not matter whether the response task taps into property (3b) (as in Exps. 2 and 5) or into property (3c) (as in Exp. 6).

Particularly important is investigating when two diagnostics that rely on distinct properties of at-issue content yield the same or different results. If we were able to unambiguously conclude that two diagnostics yield different results because they are based on different properties of at-issue content, this result would provide empirical support for the assumption that the diagnostics do not measure the same underlying concept of at-issueness. The next section follows up on this point.

4.2 Do different diagnostics measure the same underlying concept?

This section addresses research question (12b), that is, whether different diagnostics of at-issueness measure the same underlying concept. We begin again by taking stock. For NRRCs, the diagnostics implemented in Exps. 1–4 did not provide empirical support for the result of Syrett & Koev 2015: Exp. 2, that the content of sentence-final NRRCs is more at-issue than that of -medial ones; see also Snider 2017b. Should we conclude from this result that the diagnostic implemented in Syrett & Koev 2015: Exp. 2 measures a different underlying concept than the diagnostics implemented in Exps. 1–4, like Snider 2017b did? We argue that this conclusion would be premature, as the differences in the results may also be due to different interactions between the meanings of the expressions associated with the target contents and the diagnostics. For instance, we suggested in §2.3.1 that the 'asking whether' diagnostic implemented in Exp. 2 may not be suitable for detecting differences between sentence-medial and -final NRRCs, due to the fact that NRRCs contribute their own speech act. Likewise, we pointed out that there are several differences between Syrett & Koev 2015: Exp. 2 and our Exps. 3–4 (which all relied on property (3c)), which may have contributed to the differences in the results. In sum, and as suggested in the previous section, in order to claim that two diagnostics do not measure the same underlying concept, we need to establish that differences in the results are unambiguously due to the diagnostics relying on distinct properties of at-issue content and cannot be attributed to other factors, like the meaning of the expression associated with the content, the presentation of the stimuli, the items or the response task.

For the contents of the complements of clause-embedding predicates, the diagnostics implemented in Exps. 1–4 also did not give consistent results: Recall that the results of Exp. 2 ('asking whether') provided empirical support for 9 out of 10 possible pairwise differences observed in Degen & Tonhauser 2025, Exp. 5 ('asking whether') and Exp. 6 ('affirmative response'), but the results of Exps. 1 (QUD) and 4 ('yes, but') only supported 2 and 5 of the 10 possible pairwise differences, and the results of Exp. 3 (direct-dissent) supported none. We again argue that concluding from these results that the diagnostics implemented in Exps. 1, 3–4 measure a different underlying concept than those implemented in Exps. 2, 5–6 would be premature. Rather, as we hypothesized in §2.3.3, it is possible that polar interrogatives interact differently with the meanings of the expressions and the target contents than positive declaratives. In other words, the diagnostics implemented in Exps. 1–6 may measure the same underlying concept but differences emerge for the contents of the complements of clause-embedding predicates because of different interactions with polar interrogatives and positive declaratives.

In sum, with respect to research question (12b), we conclude that we are not yet in a position to determine whether different diagnostics measure the same underlying concept. If one were to observe that two diagnostics yield consistent results for any type of content they are applied to, that may be taken to provide support for the position that the two diagnostics measure the same underlying concept. But when two diagnostics don't (always) yield consistent results when applied to a broad range of contents, what is needed to determine whether the diagnostics measure the same underlying concept, or two different ones, is a deeper understanding of the workings of the diagnostic and its interaction with the meanings of expressions associated with the contents. These are important tasks for future research on at-issueness.

4.3 Methodological implications

There are several methodological implications that emerge from the results of the six experiments. A first implication is that the results of research on at-issueness may be particular to the diagnostic used in the research and may not generalize to other diagnostics. This methodological implication was already suggested in Snider 2017a; b; 2018, Koev 2018 and Korotkova 2020. In our work, the methodological implication follows straightforwardly from the observation that the results of Exps. 1–4 differed for the contents of the complements of the five clause-embedding predicates. It also follows from the observation that none of Exps. 1–4 replicated the differences between sentence-medial and -final NRRCs reported in Syrett & Koev 2015, including even Exps.3–4, which also implemented diagnostics based on property (3c).

A second methodological implication is that it is important to consider the response task with which a diagnostic is implemented. There are several pairs of experiments that suggest this methodological implication. For instance, while both our Exp. 4 and Syrett & Koev 2015: Exp. 2 implemented a dissent diagnostic, they different in the response task and yielded slightly different results. Likewise, Exps. 5 and 6 both embedded the target contents in polar interrogatives, but differed in the response task and yielded slightly different results for some pairs of clause-embedding predicates. Thus, when designing an experiment to measure at-issueness, it is important to consider not just the diagnostic, but also the response task.

A third methodological implication is that the expressions and items used in an experiment may yield different result. This implication follows from our comparison of the results of Exp. 2 ('asking whether') and Exps. 5–6. Whereas the latter two experiments provided support for all ten pairwise differences between the five clause-embedding predicates investigated in Exps. 1–4, the results of Exp. 2 only provided support for nine of the ten differences. One salient difference between Exp. 2 and Exp. 5 are the expressions investigated in the two experiments and them items. As noted in §3, Exp. 5 also consisted of a projection block, which may also have affected the results (see Degen & Tonhauser 2025).

A final methodological implication is that some diagnostics may not be suitable for some contents. Recall from §2.3.2 that we hypothesized that the content of the complement of *be right* was rated as rather unacceptable in Exp. 1 (QUD diagnostic) not because it is not-at-issue but because of the way in which the meaning of *be right* interacts with the diagnostic. Another results that suggests this methodological implication comes from the observation that Exp. 2 ('asking whether') did not show differentiation between the contents of sentence-medial and -final NRRCs. We suggested in §2.3.1 that this is due to NRRCs contributing their own speech act, which differs from the speech act of the main clause in polar interrogatives. In sum, both of these results suggest that some diagnostics may not be suitable for some contents.

5 Conclusions

At-issueness is a key concept in theoretical semantics pragmatics, but there is no consensus to date on whether different diagnostics measure the same or distinct underlying concepts, and how at-issueness is defined. This paper contributed to these overarching questions by experimentally comparing five diagnostics of at-issueness applied to the contents of NRRCs and of the complements of clause-embedding predicates. The results gave a mixed picture, as the results suggest that some of the diagnostics are not interchangeable while others are. We cautioned against taking the former result to mean that the diagnostics measure distinct underlying concepts (as done in, e.g., Snider 2017b and Koev 2018) because differences in the results of two diagnostics may also be the result of an interaction between the diagnostics and the meanings of the expressions whose contents are diagnosed. Ultimately, we proposed that further investigations are needed to determine whether different diagnostics measure the same underlying concept and the paper offered several methodological implicatures for future research on at-issueness.

Data accessibility statement

The experiments, data and R code for generating the figures and analyses of the experiments reported in this paper are available at https://anonymous.4open.science/r/at-issueness-diagnostics-232C/.

Ethics and consent

All experiments were conducted with approval from the ethics review committee of [university name redacted for review].

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Supplements

A Control stimuli in Exps. 1–4

The examples in (1)-(4) provide the two control stimuli used in each of Exps. 1-4. For the a.-examples, participants were expected to give a 'totally fits' response (Exp. 1), a 'yes' response (Exp. 2), a 'totally natural' response (Exp. 3), and a 'no' response (Exp. 4); for the b.-examples, the opposite response was expected. The numbers after each example identify the mean ratings

(Exps. 1-3) or the proportion of 'no' responses (Exp. 4) after excluding participants who did not self-identify as native speakers of American English (but before excluding participants on the basis of these controls), showing that the control stimuli worked as intended.

(1) Control stimuli in Exp. 1 (QUD diagnostic)
  a. Mary: Which course did Ava take?
     John: She took the French course. (.97)
  b. Jennifer: What does Betsy have?
     Robert: She loves dancing salsa. (.07)

(2) Control stimuli in Exp. 2 ('asking whether' diagnostic)
  a. Mary: Did Arthur take a French course?
     Question to participants: Is Mary asking whether Arthur took a French course? (.96)
  b. Robert: Does Betsy have a cat?
     Question to participants: Is Robert asking whether Betsy loves apples? (.02)

(3) Control stimuli in Exp. 3 ('direct dissent' diagnostic)
  a. Mary: Arthur took a French course.
     Lily: No, he took a Spanish course. (.87)
  b. Robert: Betsy has a cat.
     Maximilian: No, she doesn't like apples. (.05)

(4) Control stimuli in Exp. 4 ('yes, but' diagnostic)
  a. Mary: Arthur took a French course.
     Lily: Yes, but Lisa loves cats. / Yes, and he didn't take a French course. / No, he didn't take a French course. (.95)
  b. Robert: Betsy has a cat.
     Maximilian: Yes, but she is good at math. / Yes, and she loves it so much. / No, she doesn't like apples. (0)

B 20 clauses (items) used in Exps. 5–6

The contents of the following 20 clauses, which instantiated the complements of the 20 clause-embedding predicates, were investigated in Exps. 5–6:

i. Mary is pregnant.
ii. Josie went on vacation to France.
iii. Emma studied on Saturday morning.
iv. Olivia sleeps until noon.
v. Sophia got a tattoo.
vi. Mia drank 2 cocktails last night.
vii. Isabella ate a steak on Sunday.
viii. Emily bought a car yesterday.
ix. Grace visited her sister.
x. Zoe calculated the tip.
xi. Danny ate the last cupcake.
xii. Frank got a cat.
xiii. Jackson ran 10 miles.
xiv. Jayden rented a car.
xv. Tony had a drink last night.
xvi. Josh learned to ride a bike yesterday.
xvii. Owen shoveled snow last winter.
xviii. Julian dances salsa.
xix. Jon walks to work.
xx. Charley speaks Spanish.

C Control stimuli in Exps. 5–6

The control stimuli in Exps. 5–6 were the contents of the main clause polar interrogatives in (5). The non-restrictive relative clauses (NRRCs), given in parentheses in (5), were included in Exp. 6, where at-issueness was measured with an assent diagnostic. The control stimuli here consisted of two clauses (like the target stimuli), to allow the relevant speaker to assent with one of two clauses.

(5) Sentences for control stimuli in in question embedding experiments (Exps. 5–6)
  a. Do these muffins (, which are really delicious,) have blueberries in them?
  b. Does this pizza (, which I just made from scratch,) have mushrooms on it?
  c. Was Jack (, who is my long-time neighbor,) playing outside with the kids?
  d. Does Ann (, who is a local performer,) dance ballet?
  e. Were John's kids (, who are very well-behaved,) in the garage?
  f. Does Samantha (, who is really into fashion,) have a new hat?

We expected participants to give low responses on the at-issueness diagnostics for the control stimuli in (5), indicating that the main clause content is at-issue.